

Problem 1. (10 points) (Sum of a random number of random variables)

You are visiting the rainforest, but unfortunately your insect repellent has run out. As a result, at each second, a mosquito lands on your neck with probability 0.5. If one lands, with probability 0.2 it bites you, and with probability 0.8 it never bothers you, independently of other mosquitoes.

- (a) What is the expected time between successive mosquito bites? What is the variance of the time between successive mosquito bites?
- (b) In addition, a tick lands on your neck with probability 0.1. If one lands, with probability 0.7 it bites you, and with probability 0.3, it never bothers you, independently of other ticks and mosquitoes. Now, what is expected time between successive bug bites? What is the variance of the time between successive bug bites?

Problem 2. (10 points) (Covariance)

Let X and Y be independent random variables each having the uniform distribution on $[0, 1]$. Let $U = \min\{X, Y\}$ and $V = \max\{X, Y\}$.

- a) Find $\mathbf{E}[U]$.
- b) Find $\text{cov}(U, V)$.

Problem 3. (10 points) (Transforms)

Let $X_1, X_2, \dots, X_n$ be independent variables. For each $i \in \{1, 2, \dots, n\}$, X_i follows a **standard normal distribution**. Define $Y = X_1^2 + X_2^2 + \dots + X_n^2$.

- a) Find $\mathbf{E}[Y]$ and $\text{var}(Y)$.
- b) Suppose that $n = 100$. Find a good approximation for the probability that $\mathbf{P}(Y \geq 500)$. Give your answer in terms of the standard normal CDF $\Phi(x)$.

Problem 4. (10 points) (Transfer and Poisson Process)

A pizza parlor serves n different types of pizza, and is visited by a number K of customers in a given period of time. Each customer orders a single pizza, with all types of pizza being equally likely, independent of the number of other customers and the types of pizza they order.

- (a) If K is a nonnegative integer random variable with a known associated transform $M_K(s) = \mathbb{E}[e^{sK}]$. Give a formula, in terms of $M_K(\cdot)$ for the expected number of different types of pizzas ordered.
- (a) If K is a Poisson random variable with mean λ . Find the expected number of different types of pizzas ordered.

Problem 5. (15 points) (Weak law of large numbers)

Random variable X is uniformly distributed between -1.0 and 1.0 . Let $X_1, X_2, \dots$, be independent identically distributed random variables with the same distribution as X . Determine which, if any, of the following sequences (all with $i = 1, 2, \dots$) are convergent in probability.

(a) $U_i = \frac{X_1 + X_2 + \dots + X_i}{i}$

(b) $W_i = \max\{X_1, \dots, X_i\}$

(c) $V_i = X_1 \cdot X_2 \cdot \dots \cdot X_i$

Problem 6: (15 points) (Gaussian random variables)

Let X and Y be two independent variables such that $X \sim \mathcal{N}(\mu, 1)$ and $Y \sim \mathcal{N}(0, 1)$.

- (a) Find the PDF of $Z = \min(X, Y)$.
- (b) For each $t \in \mathbb{R}$, calculate $\mathbf{P}(\max(X, Y) - \min(X, Y) \geq t)$.

Give all your answers in terms of the standard normal CDF $\Phi(x)$ and its first order derivative $\Phi'(x)$.

Problem 7. (20 points) (Poisson)

A store opens at $t = 0$ and potential customers arrive in a Poisson manner at an average arrival rate of λ potential customers per hour. As long as the store is open, and independently of all other events, each particular potential customer becomes an actual customer with probability p . The store closes as soon as ten actual customers have arrived.

- a) What is the probability that the fifth potential customer to arrive becomes the third actual customer?
- b) Find the process for the arrival of actual customers and its parameter.
- c) What is the PDF and expected value for L , the duration of the interval from store opening to store closing?
- d) Considering ten actual customers have arrived, what is the probability that no two actual customers arrive within τ time units of each other?